

PBA Research Project Final Report Evaluation - Group 5

Project Title: Food Environment and Health Outcomes: Evidence on Diabetes and Obesity

Group #: 5

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I. Research Question & Objectives (30% weight)

Clarity and Specificity of Research Question(s)

Score: 5/5

The research question examines how different dimensions of food access, from retail stores and restaurants to government assistance programs, relate to adult diabetes and obesity rates across U.S. counties. The question is clear, policy-relevant, and appropriately scoped for an undergraduate project.

The framing around food environment effects on health outcomes connects to substantial public health literature. The group correctly identifies the complexity of the issue: the presence of stores does not necessarily mean access to healthy food. Citing prior research from Mississippi, the introduction notes that SNAP-authorized retailers in high-obesity counties are often convenience stores where healthy options are limited and expensive.

Feasibility and Significance of Objectives

Score: 4.5/5

The objectives are well-matched to the available data. The USDA Food Environment Atlas provides rich county-level variables spanning multiple years, enabling analysis of both cross-sectional patterns and changes over time. The group analyzes four health outcomes across different years (diabetes in 2015 and 2019; obesity in 2017 and 2022), which allows examination of whether relationships are stable or evolving.

The question has clear policy implications for food assistance programs, urban planning, and public health resource allocation.

II. Methodology (35% weight)

Appropriateness and Justification of Methods

Score: 4/5

The methodology employs two complementary analytical approaches:

1. Multiple Linear Regression (OLS): The group runs category-specific regressions (food stores, restaurants, assistance, local food systems, health infrastructure, socioeconomic factors) to examine which variable groups are associated with health outcomes, then combines variables into full models. This structure allows assessment of both category-level and combined associations. The proper use of adjusted R^2 to filter out low-explanatory categories (excluding those below 0.09) shows appropriate statistical judgment.

2. Principal Components Analysis (PCA): The group applies PCA to address multicollinearity among correlated food environment variables. PCA allows dimensionality reduction while preserving information, and the group interprets component loadings to understand what each PC represents substantively.

The group explicitly acknowledges the inability to use Difference-in-Differences: "Due to data limitations, we were unable to implement the Difference-in-Differences method (DID), as the tax change occurred in 2014, prior to the years for which health outcome data are available." This shows methodological awareness, understanding both the appeal of quasi-experimental methods and the specific data requirements they impose.

Rigour and Ethical Considerations

Score: 4.5/5

The data handling is rigorous. The group documents: (1) removal of invalid values (-8888, -9999) and conversion to NA; (2) Z-score standardization to ensure coefficient comparability; (3) exclusion of variables with excessive missing values (SUPERCPTH16, SUPERCPTH20, SPECSPTH16, SPECSPTH20); and (4) attention to multicollinearity, with specific variables removed from models when multicollinearity was detected.

The analysis presents extensive regression tables with estimates, standard errors, and p-values for each model. This transparency enables readers to assess statistical significance and effect magnitudes.

Causal language: The report maintains appropriate causal humility for most of the analysis. Words like "associated with," "correlated with," and "related to" are used correctly to describe observational relationships. However, the policy recommendations section occasionally slips into stronger language ("decrease the prevalence of diabetes and obesity") that implies causal effects not established by the cross-sectional design. This is a minor concern given the generally careful language throughout.

III. Results/Findings (15% weight)

Clarity and Presentation of Results

Score: 4.5/5

Results are presented systematically across 19 regression tables, organized by health outcome (diabetes/obesity), time period, and model type (category-specific versus combined). Each table includes all relevant statistics (estimates, standard errors, p-values, adjusted R^2). The section summaries at the end of each outcome analysis (4.1.5 Overview, 4.2.5 Overview) helpfully synthesize findings.

The PCA results include both component loadings and interpretations of what each component represents substantively (e.g., "food store accessibility" versus "socioeconomic disadvantage").

Interpretation and Analysis of Results

Score: 5/5

The interpretation demonstrates analytical maturity, particularly in handling counterintuitive findings.

The most notable example is the positive correlation between SNAP-authorized store density and higher diabetes/obesity rates. Rather than dismissing this as anomalous or claiming SNAP causes poor health outcomes, the group provides a thoughtful interpretation grounded in prior research:

"Research in the Mississippi Delta shows that the SNAP-authorized retailers in high-obesity counties are mainly convenience stores, where healthy choices are limited and much more expensive than at grocery stores. This setting essentially restricts consumers' options to cheap, high-calorie processed food, increasing risks for chronic disease."

This interpretation correctly identifies that the correlation reflects the *type* of stores authorized to accept SNAP benefits (often convenience stores with limited healthy options), not a direct causal effect of SNAP access on health.

The group also handles sign changes between category models and full models thoughtfully. This demonstrates understanding that coefficient estimates can change when controlling for additional variables, which reflects omitted variable patterns rather than true contradictions.

IV. Discussion & Conclusion (15% weight)

Discussion of Findings in Relation to Literature

Score: 4/5

The discussion engages with prior research, citing CDC studies on SNAP retailer food quality in Mississippi and research on SNAP-authorized store formats in Virginia. The connections between findings and existing scholarship strengthen the interpretation.

However, the references section includes only three academic sources. For a 25-page research report, more extensive engagement with the food environment and public health literature would strengthen the contribution. The USDA Food Environment Atlas has been used in numerous published studies; citing some of this prior work would situate the findings within the broader scholarly conversation.

Conclusions and Implications

Score: 4.5/5

The policy recommendations are specific and grounded in the analytical findings: expand specialized food stores and full-service restaurants (both negatively associated with poor health outcomes); address dollar store expansion with nutritional quality requirements; reallocate SNAP-authorized store networks toward supermarkets; strengthen local food systems (farmers' markets, farm-to-institution programs); and integrate food policy with socioeconomic development strategies.

These recommendations follow logically from the regression results. The concluding acknowledgment that "no single intervention will solve these problems" reflects appropriate understanding of the multidimensional nature of the findings.

The limitations section is thorough, discussing both methodological constraints (DiD not feasible with available data timing) and suggestions for future research (heterogeneity analysis by county characteristics, additional data on SNAP participation).

V. Writing Quality & Presentation (5% weight)

Organization and Structure

Score: 4/5

The report is well-organized with clear sections: Introduction, Research Design, Data, Methodology & Results, Recommendations, and Limitations. The variable documentation (Table of variables with descriptions) is helpful. The progression from category-specific models to combined models to PCA provides a logical analytical arc.

Clarity, Grammar, and Style

Score: 4/5

Writing is clear and professional. Technical concepts are explained accessibly. Some minor grammatical issues typical of non-native English writing but nothing that impedes comprehension.

General Comments

This is one of the strongest projects in the class. Several elements stand out:

Sophisticated interpretation: The handling of the SNAP-health correlation exemplifies how to interpret surprising findings. Rather than over-claiming ("SNAP causes obesity") or dismissing ("this must be wrong"), the group correctly interprets the pattern as reflecting the types of stores accepting SNAP, grounded in CDC research.

Methodological awareness: Explicitly acknowledging the inability to use DiD due to data timing constraints shows understanding of what different methods require. Many students would either ignore this limitation or claim stronger causal interpretation than warranted.

Comprehensive documentation: The 25 pages of analysis with 19 regression tables and PCA output demonstrates thorough work. All model specifications are fully reported with statistics enabling assessment of significance and magnitude.

Areas for Further Development

1. **Strengthen Literature Engagement:** With only three academic sources cited, the report would benefit from broader engagement with the substantial literature on food environments and health. This would help situate your findings and identify how they extend or confirm prior work.
2. **Report Confidence Intervals:** Throughout the semester, we emphasized that point estimates alone are insufficient. Presenting 95% confidence intervals alongside coefficient estimates would allow readers to assess the precision of relationships, not just whether they are statistically significant.
3. **Panel Methods:** If pursuing causal claims in future work, consider fixed effects models that exploit within-county variation over time. The cross-county comparisons in this analysis cannot rule out unobserved county characteristics that correlate with both food environment and health.
4. **Heterogeneity Analysis:** The limitations section correctly identifies this as valuable future work. Examining whether food environment effects differ by urban/rural status, poverty rates, or SNAP participation levels would add substantive depth.

Class Performance Context

Statistic	Value
Class Mean	76.5
Class Median	80
Standard Deviation	14.3
Range	35 - 92
Your Score	89

Overall Score: 89 / 100

(This will contribute 22.25 / 25 points toward your final letter grade)

Score Breakdown

Section	Weight	Score
I. Research Question & Objectives	30%	5/5, 4.5/5
II. Methodology	35%	4/5, 4.5/5
III. Results/Findings	15%	4.5/5, 5/5
IV. Discussion & Conclusion	15%	4/5, 4.5/5
V. Writing Quality & Presentation	5%	4/5, 4/5

Section	Weight	Score
TOTAL	100%	89/100